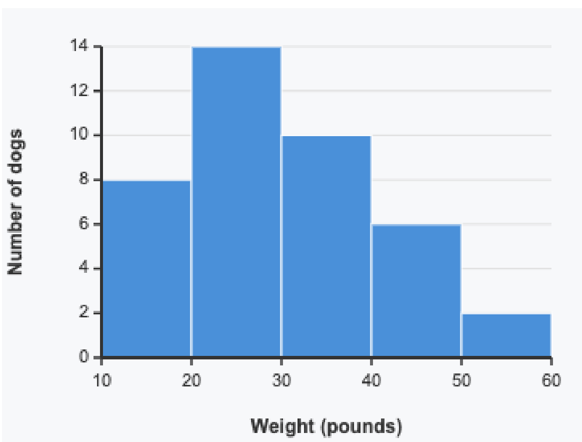
Mini Lesson Sample Script

Materials: Straightedge

1. **Interpret a histogram to answer questions about a data set.**	
The histogram shows the weights, in pounds, of dogs at a dog park. Each bar includes the left-end value but not the right-end value. For example, the first bar includes dogs that weigh 10 pounds and 15 pounds, but not 20 pounds.  a. How many dogs weigh at least 30 pounds? b. How many dogs weigh exactly 25 pounds? c. What is a typical weight for a dog at this park? *Answer: a) 18 dogs; b) Cannot be determined from the histogram (somewhere between 0 and 14 dogs); c) Answers vary. Sample response: about 25 pounds, since the tallest bar shows that the most dogs weigh between 20 and 30 pounds.*	T: This is a histogram. The x-axis shows the weights of dogs in pounds, and the y-axis shows how many dogs fall in each interval. Each bar includes the left-end value but not the right-end. So a dog weighing exactly 20 pounds would be in which bar? S: The 20 to 30 bar. T: Right. Let's look at part (a). Which bars represent dogs that weigh at least 30 pounds? S: The 30 to 40, 40 to 50, and 50 to 60 bars. T: How tall is each of those? S: 10, 6, and 2. T: So how many dogs weigh at least 30 pounds? S: 18 dogs. T: Now part (b). How many dogs weigh exactly 25 pounds?

<table>
<tr>
<td></td>
<td>S: 14 dogs? → Wait, the bar shows 14, but that's for 20 to 30 pounds.

T: Right. Can you tell from the histogram exactly how many dogs weigh 25 pounds?

S: No. It could be anywhere from 0 to 14 of those dogs.

T: Good. The histogram groups the data, so we lose the exact values. What about part (c), a typical weight?

S: Around 25 pounds, because the tallest bar is between 20 and 30 pounds.</td>
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<tr>
<td colspan="2">2. Use a frequency table to create a histogram.</td>
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<td>The frequency table shows the heights, in inches, of plants in a garden. Use the table to create a histogram on the grid.

<table>
<tr><th>Height (inches)</th><th>Frequency</th></tr>
<tr><td>0–5</td><td>3</td></tr>
<tr><td>5–10</td><td>7</td></tr>
<tr><td>10–15</td><td>12</td></tr>
<tr><td>15–20</td><td>5</td></tr>
<tr><td>20–25</td><td>2</td></tr>
</table>
</td>
<td>T: Use the frequency table to build a histogram on the grid. The x-axis is set up with intervals of 5 inches, and the y-axis counts by 2. Let's start with the first bar. How tall should the bar for 0 to 5 inches be?

S: 3.

T: I'll draw that bar from 0 to 5 on the x-axis, with a height of 3. Now you draw the bar for 5 to 10 inches. How tall is it?</td>
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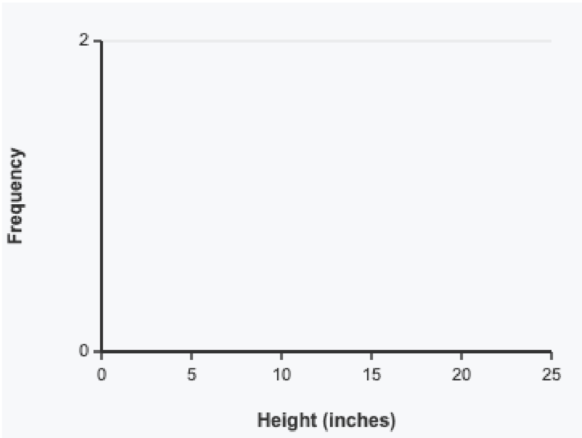
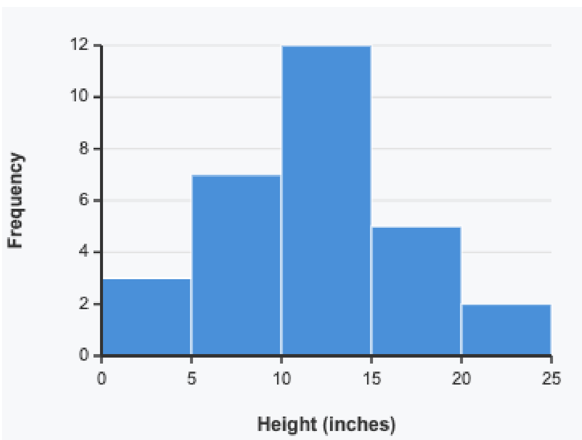
<td>

Answer:


</td>
<td>S: 7. That's between 6 and 8 on the y-axis.

T: Go ahead and finish the rest of the bars using the table.

S: (Draw.)

T: Remember the bin rule: a plant exactly 10 inches tall goes in the 10 to 15 bar, not the 5 to 10 bar.</td>
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<tr>
<td colspan="2">3. Compare a histogram and a dot plot of the same data set.</td>
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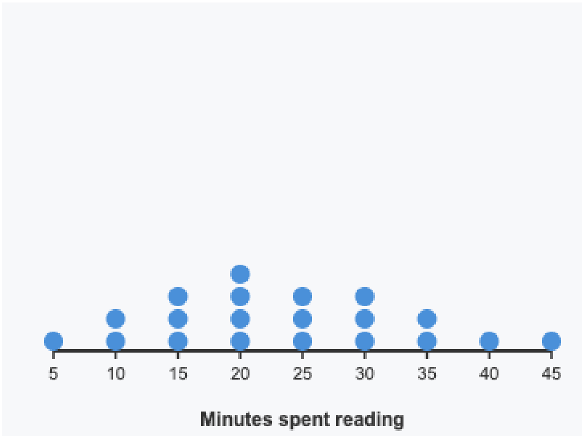
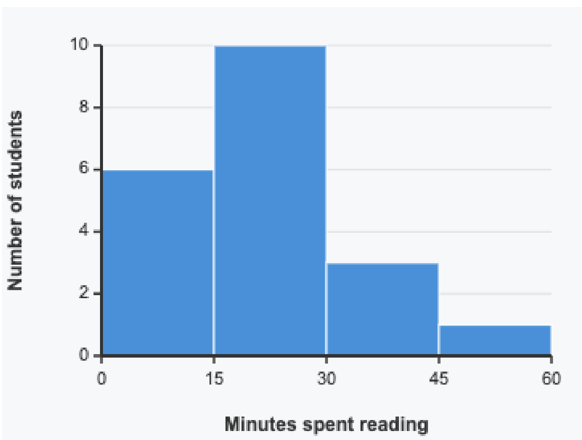
<tr>
<td>Both displays show the same data set: the number of minutes 20 students spent reading last night.</td>
<td>T: Both displays show the same data. Remember, each dot on the dot plot represents one student. For part (a), which display tells you exactly how many students read for 20 minutes?

S: The dot plot. I can count 4 dots above 20.

T: Right. Why can't the histogram tell us that?</td>
</tr>
</table>





a. Which display tells you how many students read for exactly 20 minutes? How many?

b. Which display makes it easier to see the overall shape of the data?

c. Which display would you use to find how many students read for at least 30 minutes? How many?

Answer: a) The dot plot; 4 students; b) The histogram; c) Either display works. The histogram

S: Because the bar covers 15 to 30 minutes all together.

T: For part (b), which display makes the shape of the data easier to see?

S: The histogram. The bars show where the data piles up.

T: Now part (c). How many students read for at least 30 minutes? Try both displays.

S: The histogram shows 3 plus 1, so 4. The dot plot also shows 4 dots from 30 on.

T: When would you choose a histogram over a dot plot?

S: When there's a lot of data and you want to see the overall shape, not every value.

shows 4 students; the dot plot shows 4 students (3 + 1).	